

Photo Time Classifier

Sort photos by the time they were taken

[USER MANUAL](#)

Sort a folder of photos into dated folders automatically, using the shot time recorded in each photo's EXIF data. Choose year, month, or day; photos without a shot time are set aside in **others**.

Open-source · Windows · No models, no downloads · Originals never modified

Designed by Codecider Lab

1. Overview

Photo Time Classifier is a desktop application that scans a folder of photos, reads the shot time recorded in each photo's EXIF metadata, and copies the photos into folders named after that time. You pick one grouping - year, month, or day - and the folders are created for you.

The shot time is the moment the shutter actually fired, which the camera or phone writes into the file. It is not the file's created or modified date, so copying, syncing, or re-downloading photos does not change how they are sorted.

Key features

- Sorts by EXIF shot time into **year**, **month**, or **day** folders.
- Photos with no shot time go to **others** rather than being dropped.
- Reads JPEG, HEIC/HEIF (iPhone), TIFF, PNG, BMP, and WebP files.
- Photos are **copied, never moved** - the source folder is left untouched.
- Same-named photos are kept side by side as **name_1.jpg**, **name_2.jpg**.
- Damaged photos - partially damaged or fully corrupt - are set aside in **Damaged**.
- No machine-learning models and no first-run downloads - it starts instantly.
- Responsive dark-themed interface; the app never freezes during processing.

2. Installation & First Run

Requirements

- Windows 10/11, Python 3.10 or newer.
- No GPU and no internet connection are required.

Setup

Double-click **run.bat**. On the first run it automatically creates a virtual environment, installs the dependencies, and launches the app. Later runs start the app directly.

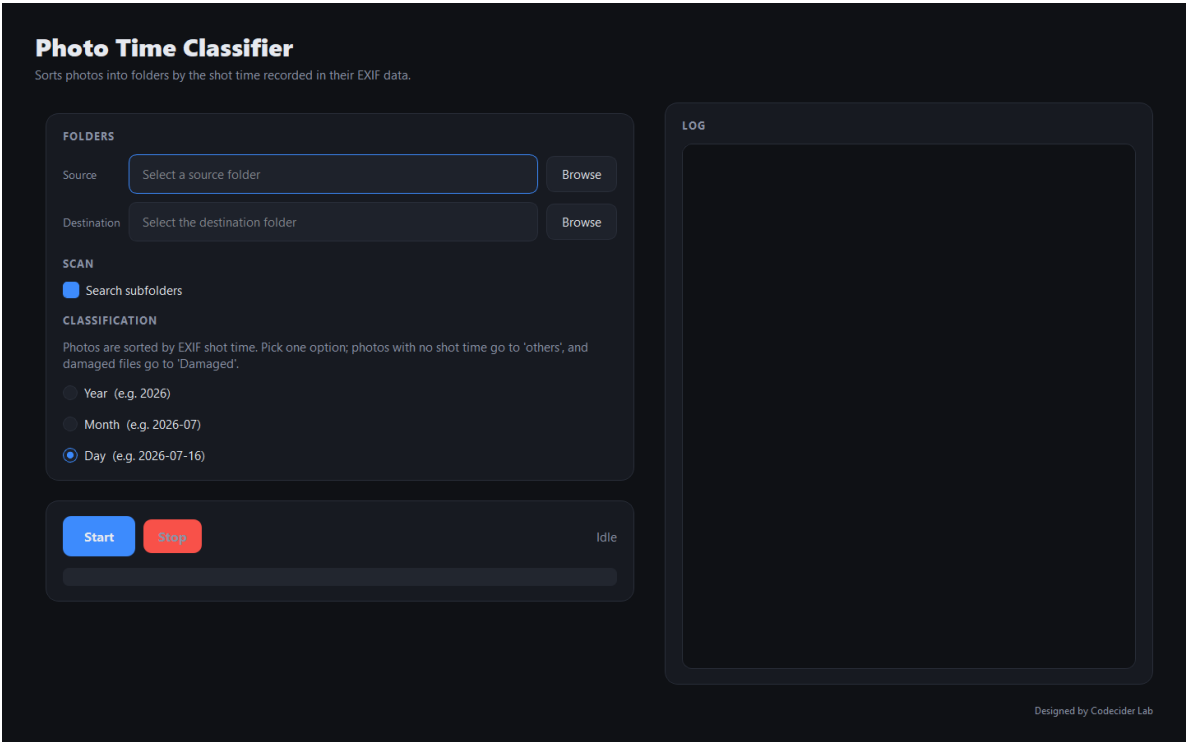
If you downloaded a pre-built release instead, run **PhotoTimeClassifier.exe** - no Python needed.

Running the tests (optional)

Double-click **test.bat** to run the unit test suite.

3. The Interface

The window has two columns. The left column holds the settings and controls; the right column shows the live log.



Area	Purpose
Folders	Choose the Source folder (input) and Destination folder (output).
Scan	Search subfolders - also collect photos from nested folders.
Classification	Pick one grouping: Year , Month , or Day .
Start / Stop	Begin or safely halt processing.
Progress bar	Shows how many photos have been processed.
Log	Per-photo results and the final summary.

4. Step-by-Step Usage

Step	Action
1	Click Browse next to Source and select the folder of photos to scan.
2	Click Browse next to Destination and select where results should be saved (must differ from Source).
3	Leave Search subfolders on to include nested folders.
4	Choose Year , Month , or Day . Only one can be selected at a time.
5	Click Start . Watch progress in the log; click Stop to halt at any time.
6	When finished, the summary line reports totals, and the dated folders are in your Destination folder.

5. Classification Options

Exactly one option is active per run - they are radio buttons, not checkboxes. The option decides how finely the photos are grouped and how the destination folders are named.

Option	Folder name	Example	Best for
Year	YYYY	2026	Long archives you want in broad buckets.
Month	YYYY-MM	2026-07	A good balance for most photo libraries.
Day	YYYY-MM-DD	2026-07-16	Trips and events, where each day matters.

Tip: run the same source folder twice with different options to compare - your originals are only ever copied, never changed.

6. Output Folder Structure

Folders are created under your Destination folder as they are needed. Photos land directly inside their dated folder; the source subfolder layout is not recreated underneath it.

Folder	Contents
2026 / 2026-07 / 2026-07-16	Photos whose EXIF shot time falls in that year, month, or day.
others	Photos with no EXIF shot time (screenshots, scans, most PNGs, stripped metadata).
Damaged	Photos that are damaged - either partially (truncated or broken pixel data) or too corrupt to open at all.

Example, with **Day** selected:

```

Destination/
  2025-12-24/ xmas.jpg
  2026-01-01/ newyear.jpg
  2026-07-16/ beach.jpg
                beach_1.jpg    <- same name, kept side by side
                sunset.jpg
  others/    screenshot.png
  Damaged/   truncated.jpg
                broken.jpg

```

7. Photos Without a Shot Time

Not every image carries EXIF data. Screenshots, scans, edited exports, downloaded images, and most PNG or BMP files have no shot time at all, and some apps strip the metadata when photos are shared. These photos are copied to **others** so nothing is silently lost, and the log shows **No shot time** for each one.

Three EXIF tags are checked, in this order:

Tag	Meaning
DateTimeOriginal	When the shutter fired. Preferred.
DateTimeDigitized	When the image was digitized (e.g. a scan).
DateTime	Last change recorded by the camera or software.

8. Damaged & Corrupt Images

Every image is checked before processing:

Condition	Handling
OK	Header and pixels decode fully → sorted by shot time.
Partially damaged	Opens but pixel data is truncated/broken → copied to Damaged .
Fully corrupt	Cannot be opened at all → copied to Damaged .

Damaged photos are separated into **Damaged** so they never mix into the dated folders. Nothing is dropped: even a file too corrupt to open is copied there, so you can inspect or recover it yourself.

9. Supported Formats

.jpg .jpeg .png .bmp .webp .tif .tiff .heic .heif

EXIF shot time is most commonly present in JPEG, HEIC, and TIFF files - the formats cameras and phones produce. Formats that carry no EXIF (typically PNG and BMP) are sorted into **others**.

10. Troubleshooting

Symptom	What to do
"Python not found"	Install Python 3.10+ and add it to PATH, then rerun run.bat.
No images found	Check the Source folder and enable Search subfolders .
Everything went to others	Those files carry no EXIF shot time - common for screenshots, scans, and PNGs, or if an app stripped the metadata when sharing.
Dates look wrong	The camera clock was wrong when the photo was taken; the app files photos exactly as the camera recorded them.
A photo is missing	Check the Damaged folder - damaged files are set aside there instead of the dated folders.
Two photos, one name	Both are kept: the second becomes name_1.jpg . Nothing is overwritten.

11. Command-Line Interface

The same pipeline is available without the GUI, for scripting and batch jobs. On Windows you can use **cli.bat** in place of **python cli.py**.

```
python cli.py SOURCE DESTINATION [options]
```

```
python cli.py ./photos ./sorted          # sort by day (default)
python cli.py ./photos ./sorted --year    # sort into 2026/
python cli.py ./photos ./sorted --month   # sort into 2026-07/
python cli.py ./photos ./sorted --day --no-recursive -q
```

Option	Description
--year	Sort into year folders (2026).
--month	Sort into year-month folders (2026-07).
--day	Sort into year-month-day folders (2026-07-16). Default.
--no-recursive	Do not scan subfolders.
-q, --quiet	Show a progress bar instead of per-image lines.

--year, **--month**, and **--day** are mutually exclusive, matching the radio buttons in the app.

Press **Ctrl+C** to stop gracefully; a summary is still printed. The exit code is **0** on success and **1** on a fatal error.

Photo Time Classifier is open-source software, released under the GNU AGPL-3.0. It uses PyQt5 (GPL-3.0), Pillow (HPND), and pi-heif (Apache-2.0). No machine-learning models are used and nothing is downloaded at first run. Designed by Codecider Lab.